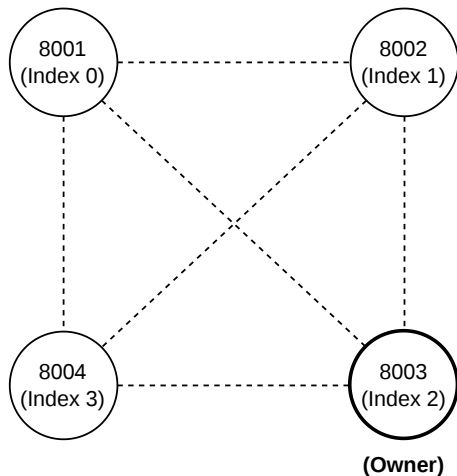


Phase 1: Stable Fully-Connected Mesh

All 4 nodes know each other. View Size (N) = 4.

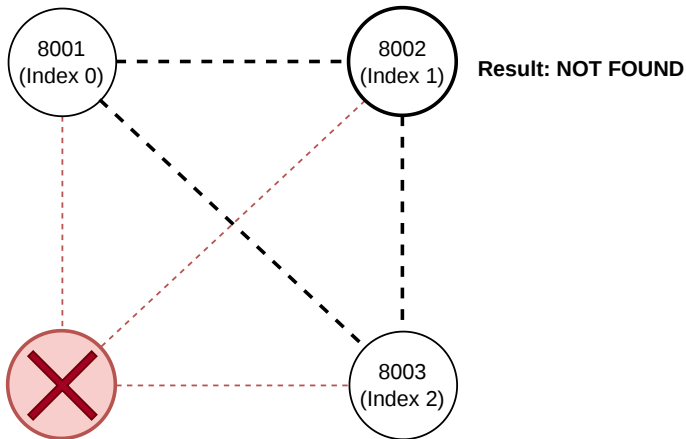


PUT "photo.jpg"

Hash("photo.jpg") = 10
 $10 \% 4 = 2$ (Node 8003)

Phase 2: Churn (A Node Disconnects)

Node 8004 crashes. Heartbeats drop it. View Size (N) = 3.



GET "photo.jpg"

Hash("photo.jpg") = 10
 $10 \% 3 = 1$ (Node 8002)

The Disaster of Naive Modulo Math

Node 8003 still has the file safely on its hard drive! But because Node 8004 crashed, the math changed. Consistent Hashing prevents this by mapping nodes to a fixed ring size (e.g., 2^{32}), making routing immune to 'N'.